

ORIGINAL ARTICLE

Total-evidence dating analysis of Osmundaceae reveals long-term extinction amid global climatic and palaeogeographical changes

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• **Background and aims** Research efforts have focused on assessing how diversification, morphological evolution and biogeographical history drive diversity across a wide array of taxonomic groups; however, their interplay remains poorly understood in ferns. Here, we conduct total-evidence dating analyses of Osmundaceae, an iconic species-poor lineage that once dominated Mesozoic forest understories. Based on dense fossil sampling and a skyline fossilized birth–death model, we assess changes in diversification rates and morphological disparity through time. Furthermore, by taking a new diffusion-based approach, we reconstruct the biogeographical history of Osmundaceae in a dynamic palaeogeographical context.

• **Methods** To perform dating analyses, we combined a novel morphological dataset of 31 characters with seven plastid markers, covering 17 extant taxa and 27 fossils. Divergence times were inferred across alternative variations of a skyline fossilized birth–death tree model. Disparity-through-time analyses were performed by reconstructing ancestral character states in the dated phylogeny inferred using the best-fitting model. Biogeographical diffusion models were reconstructed by estimating presence probabilities for each pixel across a dynamic palaeogeographical landscape and considering Earth's sphericity.

• **Key Results** Our analyses support an Early Permian radiation of fertile leptopteroids (289.32 Mya). We identified low diversification and elevated relative extinction rates, and a contraction of the morphospace, following the Toarcian Oceanic Anoxic event. High extinction rates persisted throughout the Mesozoic, delaying recovery until the Miocene. Character change rates decelerated after the Early Jurassic, coinciding with a long-term decline in lineage number and morphospace contraction. Antarctica is reconstructed as the primary centre of radiation for crown leptopteroids.

• **Conclusions** The evolutionary history of Osmundaceae is marked by prolonged high extinction, decelerated character change rates and morphospace contraction following the Toarcian Oceanic Anoxic event. Our analyses reveal that Antarctica served as the primary centre of radiation for crown leptopteroids, which diversified from an Antarctic relict population rather than via dispersal through the continent.

Key words: Antarctica, extinction, ferns, macroevolution, phylogeny.

INTRODUCTION

Diversification (speciation minus extinction) is inherently linked to biogeographical history and character evolution (Sanmartín and Meseguer, 2016; Cole and Hopkins, 2021), ultimately shaping diversity and distribution patterns through time. Speciation and extinction are known to affect morphospace occupancy through time (Puttick *et al.*, 2020). Adaptive radiation theory posits that radiation events are

defined by simultaneous high diversification and phenotypic rates, leading to rapid morphospace expansion (Hughes *et al.*, 2013). In contrast, morphospace contraction has often been explained by trait-selective extinction (Polly, 2023). Comparative studies addressing this interplay, routinely conducted in vertebrates (e.g. Slater, 2015), have either supported or rejected these trends depending on the taxonomic group. Nevertheless, with few exceptions (e.g. Matsunaga

and Smith, 2021; Flores *et al.*, 2023; Coiro and Seyfullah, 2024), macroevolutionary studies have devoted far less attention to these processes in plants. In this study, we assess the deep-time link between diversification, morphological change rates and morphospace occupancy, and biogeography in Osmundaceae, an ancient group of ferns with a rich fossil record.

Osmundaceae are leptosporangiate ferns sister to the remaining Polypodiidae (Pryer *et al.*, 2004; Smith *et al.*, 2006; Schuettpelz and Pryer, 2007; Taylor *et al.*, 2009; Rai and Graham, 2010; Escapa and Cúneo, 2012). Although their fossil record indicates that they once thrived in Mesozoic forests, their current diversity is limited to four genera and 20–23 species. Extant *Osmunda* L., *Claytosmunda* (Yatabe, Murak. & Iwats.) Metzgar & Rouhan and *Osmundastrum* Presl. occur widely in temperate and tropical regions worldwide, whereas *Plenasium* Presl. is restricted to Asia (PPG I, 2016). These genera are characterized by frond dimorphism, where sporangia-bearing frond segments and entire fronds are reduced to several degrees (Gifford and Foster, 1989). Conversely, the extant genera *Todea* Willd. ex Benth. and *Leptopteris* Presl. have laminar fertile frond segments and are sparsely distributed across southern Africa, Oceania and Asia (Kramer, 1990; PPG I, 2016). Molecular evidence has consistently recovered a ‘leptopteroid’ clade, composed of *Todea* and *Leptopteris*, sister to crown osmundeoids (*Plenasium*, *Claytosmunda* and *Osmunda*). Although affinities among extant Osmundaceae have been clarified through several molecular analyses (Smith *et al.*, 2006; Metzgar *et al.*, 2008), the phylogenetic affinities of extinct leptopteroids remain contentious.

Permineralized stems and rhizomes dominate the fossil record of Osmundaceae, although fronds are also preserved as compression–impression fossils (Miller, 1971; Carvalho *et al.*, 2013; Bomfleur *et al.*, 2017). Owing to their abundance and preservation quality, permineralized axes have often been used to conduct phylogenetic analyses (Grimm *et al.*, 2015; Bomfleur *et al.*, 2017). The subfamilies Osmundoideae and Thamnopteroidae, for instance, have been proposed based on phylogenies inferred exclusively using evidence from permineralized rhizomes (Miller, 1971; Bomfleur *et al.*, 2017). In contrast, fronds were far less studied using extensive phylogenetic approaches (Carvalho *et al.*, 2013). Sterile fronds, typically assigned to the genus *Cladophlebis*, have similarities with fronds of multiple unrelated groups of ferns (Escapa and Cúneo, 2012), hence being problematic for their use in phylogenetic analyses. Fertile fronds have been recorded from multiple localities, albeit barely included in phylogenies. In describing new leptopteroid fertile fronds, Carvalho *et al.* (2013) rendered the Eocene *Todea amissa* Carvalho *et al.* as sister to extant Australasian *Todea*, hence extending the past distribution of leptopteroids into western Gondwana. This dramatic expansion, contrasted by the restricted distribution of modern leptopteroids, raises questions about the impact of expanding the sampling of fertile fossils on biogeographical inferences within the family.

Dating key cladogenic events is a major goal in modern phylogenetics and allows understanding the macroevolutionary and biogeographical processes on a quantitative basis. In the most comprehensive dating analyses of Osmundaceae, the radiation of the family was estimated

for Late Palaeozoic–Early Triassic, and extant taxa originated in the Neogene (Grimm *et al.*, 2015). These inferences involved either a uniform tree prior or a fossilized birth–death (FBD) tree prior, and a combination of permineralized and frond fossils to calibrate internal nodes (Grimm *et al.*, 2015). The uniform tree prior is known to lead to lower marginal likelihood values and result in unrealistically ancient divergence times (see discussion by May *et al.*, 2021). In the study by Grimm *et al.* (2015), FBD estimates yielded radiation times consistent with the fossil record, although these were sensitive to changes in the data or violations of the model. In contrast to the standard FBD process (Heath *et al.*, 2014; Barido-Sottani and Morlon, 2023), the skyline FBD variation (sFBD) allows rates to shift between time bins, being more biologically realistic than the constant-rate FBD (cFBD). Despite its usage in dating analyses across taxonomic groups (e.g. Simões *et al.*, 2022), its influence on estimates of the divergence times in Osmundaceae is still underexplored.

The present study seeks to address the following question: *what macroevolutionary processes underlie the rise, decline and present poor diversity of leptopteroid Osmundaceae?* To tackle this question, we: (1) evaluate the phylogenetic affinities among fertile leptopteroid Osmundaceae from the Late Carboniferous–Eocene interval, integrating extant and fossil taxa; (2) assess macroevolutionary trends in morphology; and (3) reconstruct the biogeographical history of the family. By situating leptopteroid Osmundaceae within a deep-time macroevolutionary framework, this study examines how extinction, morphological evolution and palaeobiogeographical dynamics interacted and led to modern diversity. Overall, our results reveal a complex evolutionary history marked by an early adaptive radiation burst, long-term extinction, and speciation of relict populations. This study unveils the evolutionary trends in Osmundaceae, an iconic fern family, and provides a framework for future macroevolutionary investigations within Polypodiopsida.

MATERIALS AND METHODS

Integrating fossil evidence into extant-taxon phylogenies has typically entailed constraining node ages on the basis of putative fossil affinities (‘node dating’). One criticism of such an approach centres on the difficulty of associating fossils to a given node unambiguously (e.g. May *et al.*, 2021; Heckeberg *et al.*, 2026). Alternatively, both node ages and the position of extant and fossil taxa can be inferred jointly from combined morphological and molecular data (‘total-evidence’ approach). Here, we adopt a total-evidence approach by analysing molecular and morphological data under Bayesian inference (BI; in MrBayes; Ronquist *et al.*, 2012) and maximum parsimony (in TNT v.1.6; Goloboff and Morales, 2023). Given that we were interested in evaluating changes in diversification rates across time bins, rather than inferring a global rate, BI analyses implemented an sFBD process that allowed diversification and fossilization parameters to shift through time.

Dataset

To evaluate the phylogenetic affinities among fertile leptopteroid Osmundaceae under a total-evidence approach,

we combined the molecular dataset of Metzgar *et al.* (2008) and an extensive morphological dataset. The molecular dataset consisted of seven plastid markers for the 17 extant taxa included: *atpB*, *rbcL*, *rbcL-accD*, *rbcL-atpB*, *rps4-trnS*, *trnG-trnR* and *trnL-F* (Supplementary Data Table S1). The morphological dataset was compiled by expanding dataset from the study by Carvalho *et al.* (2013) for fertile Osmundaceae. Their dataset was originally compiled to assess the position of *Todea amissa* Carvalho *et al.*, an Eocene leptopteroid taxon assigned to the extant *Todea* clade. As such, the sampling of pre-Cenozoic leptopteroids in that matrix was limited to four species: *Todites lobatus* Naugolnykh (Late Permian), *Todites denticulatus* (Brongniart) Kasser (Jurassic), *Todites cacereii* Escapa et Cúneo (Early Jurassic) and *Todites princeps* (Presl.) Gothan (Jurassic). Here, we enhanced the sampling of fertile leptopteroids from Mesozoic strata.

In our morphological dataset, we added 22 fossils to the original dataset of Carvalho *et al.* (2013). These included frond species from the Late Carboniferous–Early Permian to the Eocene and emphasizing the inclusion of Mesozoic species of *Todites* and *Osmundopsis*. Fossils were scored from collections and literature of reference (Supplementary Data Table S1). The final dataset comprised 44 taxa, 40 belonging to Osmundaceae: *Todea* Willd. ex Bernh. (3 species), *Leptopteris* Presl. (3 species), *Todites* Seward (16 species), *Osmundastrum* Presl. (1 species), *Cacumen* Cantrill et Webb (1 species), *Osmundopsis* Harris (9 species), *Osmunda* L. (3 species), *Plenasium* Presl. (3 species) and *Claytosmunda* (Yatabe, Murak. & Iwats.) Metzgar & Rouhan (1 species; see list of species in Supplementary Data Table S1). This represented a complete sampling at the genus level and 62 % of the extant species. The outgroup comprised four extant species of Gleicheniales: *Matonia pectinata* R. Br., *Gleichenella pectinata* (Willd.) Ching, *Dipteris conjugata* Reinw. and *Diplopterygium bancroftii* (Hook.) A.R. Sm. (Supplementary Data Table S1).

Sampling of morphological characters comprised 31 characters; 16 were drawn from vegetative fronds and 15 from fertile fronds. We scored two new characters, and six characters were considered additive to reflect differential homology between states. The scoring of eight cells in the original dataset of Carvalho *et al.* (2013) was reviewed and corrected. In comparison to the dataset of Carvalho *et al.* (2013), our matrix resulted in 1062 scored cells, representing a 166.3 % increase in data volume. Topologies inferred under parsimony were rooted in *Dipteris conjugata*. To reduce the possibility of inferring impossibly old ages caused by long-branch attraction or ‘deep-root attraction’ (Zhang *et al.*, 2023), we excluded the outgroup from clock analyses under BI. The morphological dataset, including the definition of characters, is available from MorphoBank (project 6170). The character list is available under Appendix S1.

Inferences under maximum parsimony

Under the parsimony criterion, tree searches were conducted using extended implied weighting, by weighting characters against their homoplasy while assuming that missing entries contribute to half of the observed homoplasy (see Goloboff, 2014). In TNT, to distinguish missing entries

(‘?’) from inapplicable characters, the latter were scored using an asterisk (‘*’) instead of the typical dash (‘-’). Under implied weighting, characters are downweighted by using a concave function of homoplasy whose shape is defined by a constant (k), where lower values weigh more aggressively against homoplasy. Given that there is no approach to select a single concavity value, we explored three concavities ($k5$, $k10$ and $k15$). Tree searches were performed using the 25 cycles of tree drifting, sectorial searches, and five cycles of tree fusion implemented under the ‘new technologies’ algorithms in TNT (Goloboff, 1999), using 100 initial replicates and finishing the search when the optimal score was hit ten times. Multiple optimal trees, when recovered, were summarized in strict consensus trees.

Data partitioning under Bayesian inference

In clock and non-clock BI, we evaluated two partitioning schemes for the morphological data: partitioning by the number of character states (‘state partitioning’) and further by character source (i.e. vegetative or fertile characters, ‘source partitioning’). The first, relatively common in empirical analyses (Close *et al.*, 2015; Gavryushkina *et al.*, 2017; Heckeberg *et al.*, 2026), led to three partitions: binary characters, three-state characters and four-state characters. The second, less common in botanical studies, concluded in five morphological partitions: binary vegetative, three-state vegetative, four-state vegetative, binary fertile and three-state fertile. Both partitioning schemes thus guarantee that characters within a partition shared a common Q matrix (Heckeberg *et al.*, 2026). The substitution rate was modelled through the Mk model, considering variable scoring while a gamma distribution was used for the among-character rate variation. The best-fitting partitioning scheme for the molecular data was estimated with PartitionFinder2 using the corrected Akaike information criterion (Lanfear *et al.*, 2017). All subsequent parameters were unlinked across these partitions under both clock and non-clock BI analyses.

Skyline fossilized birth–death model settings

In clock analyses implementing the fossilized birth–death model, terminals are sampled along the process in different ways. We evaluated two sampling strategies implemented under the sFBD process in MrBayes: ‘random’ and ‘diversity’. The ‘random’ and ‘diversity’ strategies differ in that extant taxa are sampled at random (uniformly) or such that they maximize diversity of the lineages included, respectively (Höhna *et al.*, 2011). For these sampling strategies, net diversification ($\lambda - \mu$), relative extinction (turnover; μ/λ) and fossilization sampling [$\Psi/(\Psi + \mu)$] rates are allowed to shift across time bins.

In our sFBD, net diversification, relative extinction and relative fossilization rates were allowed to shift at three time breaks based on geological and climatic events deemed influential on plant communities: Pliensbachian–Toarcian Oceanic Anoxic event (193 Mya; Caruthers *et al.*, 2013), initial opening of the Drake Passage and thermal isolation of Antarctica (49 Mya; Scher and Martin, 2006; Vincze *et al.*, 2021), and the peak of the Middle Miocene Climatic

Optimum (16 Mya; [Methner et al., 2020](#)). We set an exponential prior with a mean of one on the net speciation rate, and uniform Beta(1,1) priors on both the relative extinction rate and the fossilization proportion. In our dataset, ~62 % of the extant species are sampled. Therefore, to account for incomplete extant taxon sampling, the ‘sampleprob’ (rho) parameter was set to 0.62 for all the clock analyses.

Two relaxed clock models were evaluated herein: independent gamma rate (IGR) and continuous autocorrelated clock (Tk02; [Thorne and Kishino, 2002](#)). For both clocks, the variance was shaped with an exponential distribution of base 20. As in previous studies (e.g. [Flores et al., 2023](#)), the clock rate was derived from the tree height inferred from the non-clock analyses relative to the tree age: 0.01881 (5.645/300) and 0.02717 (8.15/300) for ‘state’ and ‘source’ partitioning, respectively. Clock rates were then sampled from a lognormal distribution using a natural scaled mean [‘state partitioning’: $\ln(0.01881) = -3.973$; ‘source partitioning’: $\ln(0.02717) = -3.606$] and a standard deviation of 0.5. This creates a right-skewed distribution, which weighs the tree ages around 300 Mya (agreeing with the Late Carboniferous–Early Permian fossil record of the family). For each fossil taxon, the age was assigned a uniform prior bracketed by its minimum and maximum possible ages based on their stratigraphic range.

Altogether, eight sFBD model variations were assessed for their fit to the data by means of their marginal likelihood and Bayes factors as inferred in stepping-stone (SS) analyses. The eight models combined two partitioning schemes (‘state partitioning’ and ‘source partitioning’), two relaxed clock models (‘IGR’ and ‘Tk02’) and two sampling strategies (‘diversity’ and ‘random’). Systematic conclusions and downstream analyses were based on the dated topology inferred from the best-fitting model as evaluated in SS analyses.

Inferences implemented eight Markov chain Monte Carlo chains in eight parallel runs with 200×10^6 generations, sampling every 1000th generation. Initial analyses showed poor convergence after 200×10^6 generations in model variations that implemented the Tk02 clock. To improve convergence, we constrained the backbone among extant taxa as supported by non-clock analyses (i.e. topological constraints only; node ages were not constrained). Convergence was diagnosed with the average standard deviation of split frequencies (ASDF < 0.005), potential scale reduction factors (PSRF ~1.0), effective sampling size (ESS > 200), and visually on Tracer v.1.7 ([Rambaut et al., 2018](#)). Topologies under BI were summarized by considering all the compatible topologies after the burn-in phase (‘allcomp’ option; 25 %). BI analyses were performed by using the computational resources from ‘UNC Supercómputo (CCAD)’, which is part of Sistema Nacional de Computación de Alto Desempeño (SNCAD), Argentina.

Character change rates and disparity through time

To evaluate whether inferred extinction rate patterns were accompanied by changes in morphological evolutionary trends, we next evaluated character change rates and changes in disparity through time (DTT) based on the tree inferred from the best-fitting sFBD model. Per-branch character change rates in each morphological partition were compared with the mean rate in the R package ‘EvoPhylo’

([Simões and Pierce, 2021](#); [Simões et al., 2023](#)), using three categories: slower than the lower 95 % confidence interval (CI); within lower and upper 95 % CI; and faster than the upper 95 % CI.

Three DTT metrics were explored here to capture complementary properties of the morphospace ([Guillerme and Cooper, 2018](#); metrics further checked with ‘moms’, see <https://tguillerme.shinyapps.io/moms/>): sum of ranges (size of the morphospace); average displacements (relative position of a clade relative to morphospace centre); and average minimum spanning tree distance (clustering of individual observations or density). DTT metrics were computed using the R package *disparity* ([Guillerme, 2018](#)) by considering the ancestral character states at the inner nodes of the dated phylogeny, as inferred under maximum likelihood in the package *Claddis* ([Lloyd, 2016](#)). Using the default settings in *Claddis*, distance and ordination matrices were subsequently computed from the complete morphological dataset upon which the DTT metrics were computed. To evaluate whether disparity patterns differed between fertile and vegetative characters, DTT metrics were also calculated for vegetative-only and fertile-only character sets. To consider effects related to the character coding (see [Voje et al., 2023](#)), DTT analyses were repeated after subdividing the entire matrix ordinal (i.e. no quantitative transformational) and nominal (e.g. neomorphic) characters.

Biogeographical analyses under a spheric diffusion process

To determine whether diversification shifts and morphological evolution were geographically structured, we reconstructed the biogeographical history of Osmundaceae. Biogeographical reconstructions are often conducted by using predefined areas ([Buerki et al., 2013](#); [de la Estrella et al., 2019](#)) and by modelling cladogenetic and extinction events ([Ronquist, 1997](#); [Ree et al., 2005](#); [Goldberg et al., 2011](#); [Matzke, 2014](#); [Landis et al., 2022](#)). However, predefined areas may include allopatric taxa with different histories. To overcome this issue, we adopted a geographically explicit approach that models diffusion across a dynamic palaeogeography, as implemented in *PhyGeo* ([Arias, 2024](#)). In contrast to other diffusion-based approaches, *PhyGeo* explicitly accounts for Earth sphericity, thereby avoiding the spatial distortions inherent to planar representations.

PhyGeo uses a time-calibrated phylogeny, georeferenced points, and palaeogeographical and palaeolandscape models. Here, we used the tree inferred from the best-fitting model, with the e360 pixelated palaeogeographical and palaeolandscape reconstructions of [Müller et al. \(2022\)](#) and [Cao et al. \(2017\)](#), respectively, using 5 Myr time bins. Pixel-specific landscape weights were assigned as 1.0 for emerging land (both lowlands and highlands) and prohibitive values (0.0) for submerged (epicontinental and open oceans) and ice-sheet-covered areas.

Georeferenced occurrence data for extant taxa were obtained from GBIF, retaining only specimen-based records that matched taxon names in the phylogeny and whose distributions are consistent with the records of the ‘Tropicos’ collection database (49 613 curated points). For fossils, georeferenced points were extracted from the literature of reference, using either the exact published coordinates

(when available) or approximating the coordinates based on the fossiliferous site location. Fossil georeferenced points were subsequently rotated into palaeocoordinates according to the time-calibrated phylogeny and the corresponding palaeogeographical model (76 curated points; [Supplementary Data Table S1](#)).

The spherical diffusion process is governed by a concentration parameter (λ) estimated from the dated topology, where lower values correspond to faster diffusion (with $\lambda=0$ yielding a uniform distribution over the sphere), obtained using a hill-climbing optimization algorithm. Ancestral range maps were inferred by estimating conditional likelihoods for each pixel and performing stochastic mapping using 1000 particles. The resulting range maps are then smoothed through a kernel density estimation based on the spherical normal. The final ancestral range maps depict pixels within 95 % of the cumulative kernel density.

Distances were calculated as great-circle distances between the starting and ending pixels for each temporal segment along a branch, and particle speed was expressed as kilometres traversed per million years. Distances were summed across all time segments encompassed by each branch. To assess whether the inferred speed differs from expectations under the diffusion model, 1000 particles were simulated using the estimated λ and the set temporal segmentation (5 Myr). We report the proportion of observed particles moving faster than the 95th percentile or slower than the 5th percentile of the simulated distributions.

RESULTS

Phylogenetic relationships under non-clock analyses

Under both parsimony and BI, Osmundaceae was monophyletic ([Supplementary Data Figs S1 and S2](#)). Across the analyses, *Todea*, *Leptopteris* and *Todites* constituted a leptopteroid clade (*sensu* [Carvalho et al., 2013](#); [Fig. 1A](#); [Supplementary Data Figs S1 and S2](#)). Under implied weighting parsimony, a single optimal topology is recovered under each concavity ([Supplementary Data Fig. S1](#)). Under the strongest concavity (k_5), *Osmundopsis sturii* (Raciborsky) Harris was sister to the remaining genera of Osmundaceae, whereas under weaker concavities crown osmundeoids were sister to *Osmundopsis*, *Cacumen* and the leptopteroid genera ([Supplementary Data Fig. S1](#)). The genus *Todea* was sister to a clade comprising extant *Leptopteris* and a monophyletic *Todites* under k_5 ([Supplementary Data Fig. S1](#)). Conversely, *Todites* was sister to a clade comprising *Todea* and *Leptopteris* under weaker concavities (k_{10} – k_{15} ; [Supplementary Data Fig. S1](#)).

In non-clock Bayesian topologies, both ‘state’ and ‘source’ partitioning strategies resulted in the same consensus topology ([Supplementary Data Fig. S1](#)). *Osmundastrum cinnamomeum* was the single taxon sister to the remaining Osmundaceae, in agreement with previous analyses ([Metzgar et al., 2008](#); [Carvalho et al., 2013](#); [Grimm et al., 2015](#)). Crown osmundeoids were rendered sister to a leptopteroid clade comprised by *Todea*, *Todites* and *Leptopteris* ([Supplementary Data Fig. S1](#)). In the leptopteroid clade, *Cacumen expansa* was sister to a monophyletic *Osmundopsis*, together inferred as sister to *Todites* ([Supplementary Data Fig. S1](#)).

Phylogenetic relationships, diversification trends and divergence times under sFBD

In clock analyses, the model involving ‘source partitioning’ + IGR + ‘random’ (M2) outperformed the remaining models. Relative to the second best-fitting model (‘state partitioning’ + IGR + ‘diversity’, M3), the favoured model had a difference of 133.66 log units (Bayes Factor: 267.32; [Supplementary Data Table S2](#)). Model fit was mostly driven by the clock model, with IGR being favoured consistently. For example, under ‘state partitioning’ and ‘diversity sampling’, IGR (M3) was favoured over Tk02 (M7) by 266.62 log units (2lnBF = 533.24; [Supplementary Data Table S2](#)). However, the performance of partitioning and sampling strategies was determined by interactions between these parameters. In comparison to partitioning by the number of character states, further partitioning by ‘source’ led to better-fitting models whenever sampling of extant taxa in the sFBD process was ‘random’ (e.g. M5 vs. M6: 2lnBF = 33.6; [Supplementary Data Table S2](#)). The effect of the sampling strategy also depended on other settings: random sampling was favoured under ‘source partitioning’ + TK02 (e.g. M5 vs. M7: 2lnBF = 453.08), whereas diversity sampling was favoured under ‘state partitioning’ + IGR (e.g. M3 vs. M5: 2lnBF = 351.10; [Supplementary Data Table S2](#)). These results suggest that, for this dataset, clock model and character partitioning are both more influential than the specific sampling strategy within the skyline FBD process.

Under our best-fitting model (‘source partitioning’ + IGR + ‘random’, M2; [Supplementary Data Table S2](#)), Osmundaceae diverged in the Late Carboniferous–Permian [300.34 Mya, Highest Posterior Density (HPD) 298.89–303.26 Mya; clade I; [Fig. 1A](#); [Table 1](#)], consistent with the fossil record ([Miller, 1971](#); [Bomfleur et al., 2017](#)). This estimate pre-dates Middle Triassic divergence times inferred under cFBD analyses (243 Mya) and the Toarcian origin (185 Mya) recovered in total-evidence dating (TED) analyses combining permineralized rhizomes and molecular data ([Grimm et al., 2015](#)). Under this model, *Osmundopsis* was rendered non-monophyletic, with *Osmundopsis zunigai* being sister to *Cacumen expansa* and crown osmundeoids (*Osmunda*, *Plenasium* and *Claytosmunda*), defining an ‘osmundeoid’ clade that diverged in the Permian (275 Mya, HPD 255.61–297.03 Mya, clade II; [Fig. 1A](#); [Table 1](#)). Crown osmundeoids radiated in the Late Jurassic (152.87 Mya, HPD 134.44–171.3 Mya, clade VII; [Fig. 1A](#); [Table 1](#)); contrasting with previous analyses that inferred an Early Cretaceous origin for crown osmundeoids (143–133 Mya; [Table 1](#); [Grimm et al., 2015](#)). The favoured model also inferred a leptopteroid clade, including a non-monophyletic *Todites*, diverging in the Permian (289.32 Ma, HPD 279.35–298.84 Mya, clade III, [Fig. 1A](#); [Table 1](#)). In this clade, crown group leptopteroids (i.e. *Todea amissa* + extant *Todea* + *Leptopteris*) radiation was inferred in the Early Cretaceous (135 Mya, HPD 114.6–155.73 Mya, clade IV; [Fig. 1A](#); [Table 1](#)). The genera *Todea* and *Leptopteris* themselves radiated after the Cretaceous–Palaeogene mass extinction (clades V and VI; [Fig. 1A](#); [Table 1](#)).

Although relative extinction rate distributions were rather broad in the last two time bins (TBiii and TBiv), extinction concentrated at higher values in the first time bins: ~0.625

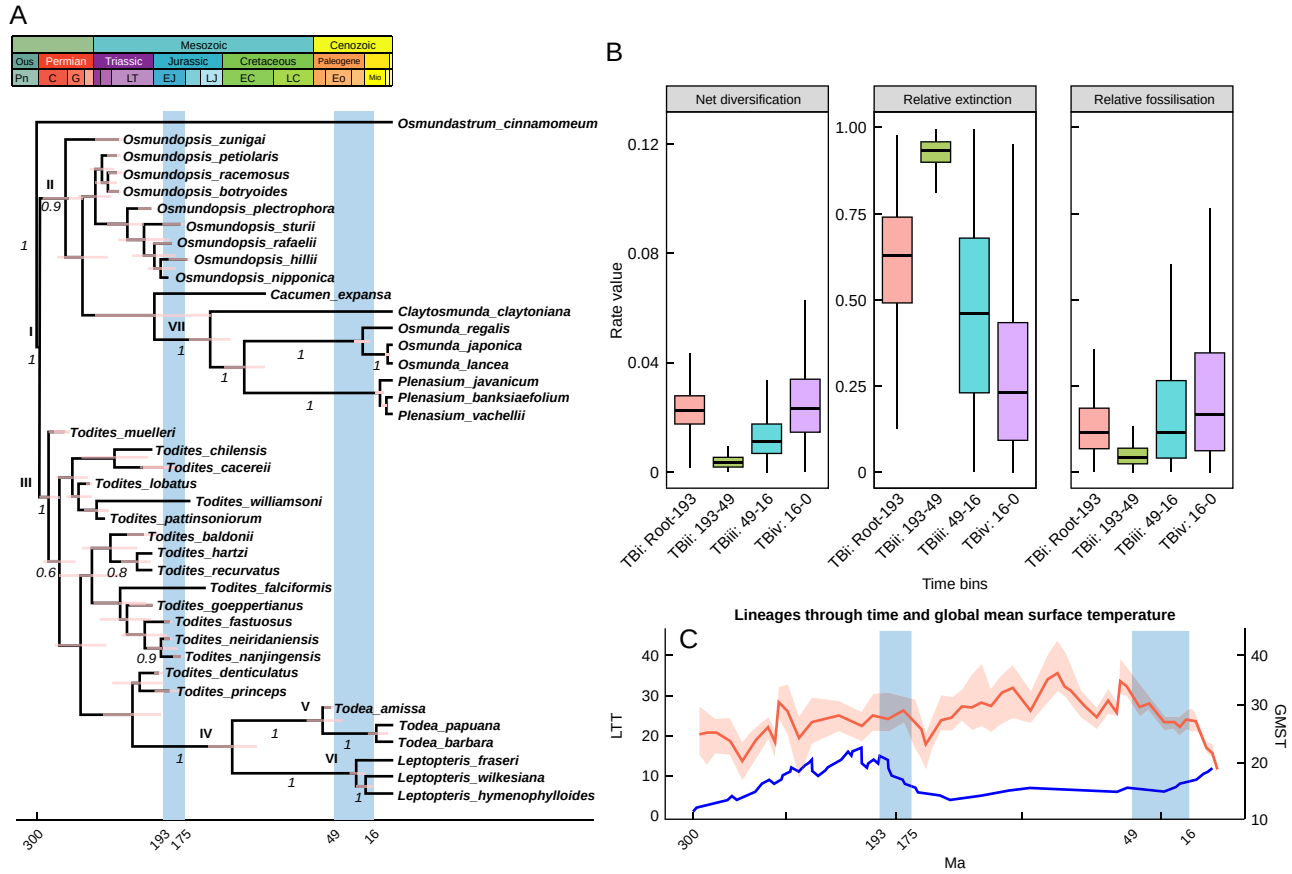


FIG. 1. (A) Dated phylogeny under the best-fitting model ('source partitioning' + IGR + 'random'; [Supplementary Data Table S2](#)). Posterior probability values >0.5 are indicated below branches. Horizontal red bars indicate HPD ranges. Ages in the bottom time scale are given in million years ago. Roman numerals indicate clades of reference for comparisons: Osmundaceae (I), osmundoid clade (II), leptopteroid clade (III), crown leptopteroids (IV), *Todea* (V), *Leptopteris* (VI) and crown osmundoids (VII; see [Table 1](#)). (B) Diversification ($\lambda - \mu$), relative extinction (μ/λ) and fossilization [$\Psi/(\Psi + \mu)$] rates across four time bins (TB) set in the skyline FBD: root time–193 (TBi), 193–49 (TBii), 49–16 (TBiii) and 16–0 (TBiv). (C) Global mean surface temperature (GMST, with 95 % CI in light red) and smoothed lineage-through-time (LTT, blue) plots across geological time. Vertical light-blue rectangles indicate the entire span of the Toarcian Oceanic Anoxic (TOA) event (193–175 Mya; [Caruthers *et al.*, 2013](#)) and the initial opening of the Drake Passage culminating in the drift of Antarctica (49–16 Mya; [Scher and Martin, 2006](#); [Vincze *et al.*, 2021](#)). Palaeotemperature was taken from [Judd *et al.* \(2024\)](#).

Table 1. Median divergence times (in million years ago) estimated for seven clades of reference (roman numerals in [Fig. 1](#)) under the best-fitting model. The corresponding 95 % Highest Posterior Density (HPD) intervals are shown in brackets. When available, comparative estimates from previous studies are provided; a dash ('–') indicates cases where no comparable estimate exists.

Clade of reference	Median inferred age (95 % HPD) (Mya)		
	Grimm <i>et al.</i> (2015)	This study	
	Total evidence (uniform tree prior)	cFBD	sFBD
(I) Osmundaceae	243 (264–233)	185 (187–183)	300.34 (303.26–298.89)
(II) Osmundoid clade	–	–	275.48 (297.03–255.61)
(III) Leptopteroid clade	–	–	289.32 (298.84–279.35)
(IV) Crown leptopteroids	113 (150–66)	116 (132–100)	135.16 (155.33–114.59)
(V) <i>Todea</i>	–	13 (23–4)	60.31 (79.81–48.32)
(VI) <i>Leptopteris</i>	49 (105–9)	27 (34–21)	30.35 (38.52–22.62)
(VII) Crown osmundoids	143 (171–110)	133 (150–119)	152.87 (171.3–134.44)

(TBi: root time–193 Mya) and ~ 0.9 (TBii: 193–49 Mya; [Fig. 1B](#)). The net diversification rates concentrated at low values (<0.04) throughout Osmundaceae history, although

a subtle increase was inferred after 16 Mya ([Fig. 1B](#)). Relative fossilization rates were likewise concentrated at low values across time bins ([Fig. 1B](#)), suggesting the

presence of ‘ghost lineages’ in the phylogeny of fertile Osmundaceae. In parallel, lineages-through-time (LTT) slope declined between 193 and 49 Mya along with a sharp rise in global mean surface temperature (Fig. 1C). A posterior increase in LTT was concomitant with the reduction of global mean surface temperature initiated by the opening of the Drake Passage at 49 Mya and afterwards by the Middle Miocene Climate Transition after 16 Mya (Fig. 1C).

Character change rate and DTT

As inferred under the best-fitting model, per-branch character change rates in vegetative characters remained within the 95 % CI boundary throughout most of the phylogeny (Fig. 2A). Binary vegetative characters accelerated above the upper 95 % CI at the radiation of the osmundeoid clade (‘SO’ + ‘CO’; Fig. 2A), whereas rates dropped below the lower 95 % CI in crown osmundeoids (‘CO’; Fig. 2A). Three-state vegetative characters, however, accelerated above the upper 95 % CI only at the radiation of crown osmundeoids (‘CO’) and crown leptopteroids (‘CL’; Fig. 2A). Rates in binary fertile characters accelerated above the lower 95 % CI mostly within stem and crown osmundeoids (‘SO’ and ‘CO’), whereas rates within stem and crown leptopteroids (‘SL’ and ‘CL’) were generally below the lower 95 % CI (Fig. 2A). Three-state fertile characters had rates above the upper 95 % CI mostly at the early radiations (>200 Mya) of both osmundeoids and leptopteroids.

Disparity analyses revealed drastic changes in morphospace occupancy across time. In assessing disparity as the sum of ranges (size of the morphospace), three stages are identified: (1) a fast increase in the interval 279–200 Mya; (2) a fast reduction in the interval 200–160 Mya; and (3) a steady (yet slight) recovery towards the present day (Fig. 2B). The other two metrics, conversely, were more stable in their trend through time. The average minimum spanning tree (morphospace density) remained relatively constant through time, albeit with minor fluctuations at ~100 Mya (Fig. 2B). The average clade position (displacement of clades relative to the centre of the occupied morphospace) persisted low and constant through time, with minor changes at ~100 Mya (Fig. 2C). These patterns were observed consistently for morphology, regardless of the data source (fertile vs. sterile) or coding strategy (Supplementary Data Fig. S3).

Biogeographical history

The concentration parameter (λ) estimated for the dated topology was 35.155 (logLikelihood = -385.643), which represents a mildly fast diffusion process based on simulations (Arias, 2024). The inferred average distance for the complete tree was 1 196 387.778 km, and the average speed of a particle was 537.912 km Myr⁻¹ (0.084 radians Myr⁻¹). Ancestral ranges inferred for deep-time branches led to reconstructions with higher uncertainty in comparison to younger branches (visually represented as geographically spread peaks with high probability; Arias, 2024). Although uncertainty commonly increases with time in deep-time inferences, uncertainty in diffusion models can be also explained by disjunct distributions and extensive widespread

distributions alike (Bouckaert *et al.*, 2012; Johansson *et al.*, 2018; Arias, 2024).

Osmundaceae and the osmundeoid clade were likely to be widespread across a Pangea continent in the interval 300–275 Mya (Fig. 3A; Supplementary Data Video S1). From this widespread distribution, crown osmundeoids diffused to northern latitudes (Fig. 3B), with extant *Osmunda* primarily spreading across Laurasia (Supplementary Data Video S2) and extant *Plenasium* concentrating towards Tropical Asia (Fig. 3C; Supplementary Data Video S3).

In the reconstructed diffusion models, Antarctica was seen to play an important role in the diffusion of leptopteroid taxa (Fig. 3D–F; Supplementary Data Videos S4–S6). The leptopteroid clade was initially concentrated at higher latitudes towards Laurasia in the interval 300–275 Mya (Fig. 3D). As opposed to the osmundeoid clade, the leptopteroid clade is characterized by a diffusion from northern to southern latitudes, with crown leptopteroids diffusing from Laurasia and spreading across eastern Gondwana (Fig. 3E, F; Supplementary Data Videos S4–S6). Crown leptopteroids diffused along this north–south path with an average speed of 536.222 km Myr⁻¹ (0.084 radians Myr⁻¹), nearly identical to crown osmundeoids across Laurasia (532.239 km Myr⁻¹; 0.084 radians Myr⁻¹; Supplementary Data Table S3). Amongst Osmundaceae, stem osmundeoids and stem leptopteroids diffused faster than younger branches, with the slowest terminal branches being those of extant leptopteroids (Supplementary Data Table S3).

In the following section, we discuss the biogeographical history of major clades of interest (Fig. 1). However, PhyGeo estimates ancestral ranges for all branches in the tree. The complete set of images is available at GitHub: <https://github.com/jorgerfl/osmunda26>.

DISCUSSION

Our analyses infer novel relationships and divergence times for leptopteroid Osmundaceae and reveal extended phases of extinction between the Jurassic and the Eocene–Miocene. High-extinction phases took place in periods of high global temperature, morphospace contraction and active diffusion across eastern Gondwana. Together, these results clarify the macroevolutionary history of Osmundaceae and highlight the impact of climate changes on the diversity of the group.

Phylogeny and radiation times of fertile Osmundaceae under sFBD

Our best-fitting dated analysis yielded both novel relationships and divergence times. The radiation of a well-supported osmundeoid clade was inferred in the Permian (Fig. 1A; Table 1), including *Cacumen expansa* and a non-monophyletic *Osmundopsis* as the stem group to crown osmundeoids. Our analysis also placed the radiation of crown osmundeoids in the Late Jurassic (152.87 Mya, HPD 134.44–171.3 Mya, clade VII; Fig. 1A; Table 1). Likewise, in addition to the mutually monophyletic *Todea* and *Leptopteris*, the Permian leptopteroid clade includes a non-monophyletic *Todites*; with *Todites muelleri* being sister to the remaining leptopteroids (clade III; Fig. 1). In comparison to the previous

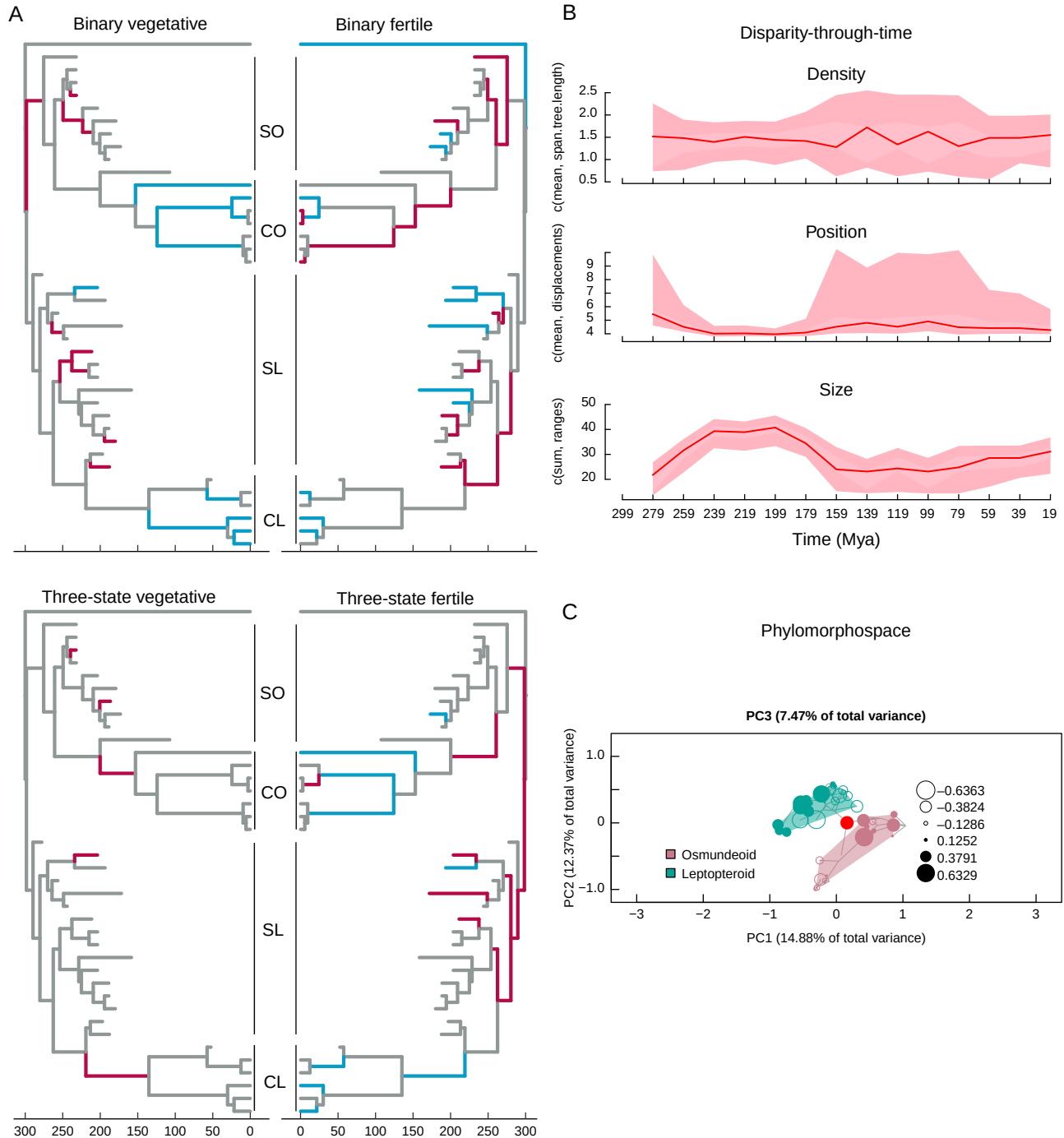


FIG. 2. (A) Dated phylogenies, with branches coloured according to change rates in four data partitions (binary vegetative, binary fertile, three-state vegetative and three-state fertile). Colour code indicates: 'slower than the lower 95 % CI' (light blue); 'within lower and upper 95 % CI' (grey); and 'faster than the upper 95 % CI' (red). Main groups of interest are indicated as: 'SO' (stem osmundoids); 'CO' (crown osmundoids); 'SL' (stem leptopteroids); and 'CL' (crown leptopteroids). (B) Disparity-through-time plots as constructed based on the entire morphological dataset and estimating three morphospace properties: density (average minimum spanning tree distance); position (average displacement relative to the centre); and size (sum of ranges). The red line indicates the median trend, and lighter red colours represent the 50 and 90 % CI. (C) Phylomorphospace occupancy, with three axes explaining 34.72 % of the variance (the third axis is given as circle sizes). Osmundoid and leptopteroid clades are highlighted in light red and light green, respectively.

inferences of fertile fronds (Carvalho *et al.*, 2013), a *Todites* + *Leptopteris* relationship is contradicted by our analyses across both maximum parsimony and BI approaches (Fig. 1; Supplementary Data Fig. S1). The radiation of crown

leptopteroids is herein inferred in the Early Cretaceous (135.16 Mya) and that of extant leptopteroids in the interval Palaeocene–Oligocene (60.31–30.35 Mya; Fig. 1; Table 1). The radiation of the genus *Todea* (i.e. including the fossil

T. amissa) is here inferred at Palaeocene (clade V; Fig. 1), differing from previous analyses (Table 1).

In addition to recovering non-monophyletic *Osmundopsis* and *Todites*, our analyses also inferred novel placements for the species of these genera relative to other taxa (Fig. 1; Supplementary Data Figs S1 and S2). Phipps *et al.* (1998: p. 894) speculated that *Osmundopsis* could have ‘given rise to *Todites*’ and that *Todites* could have been ‘ancestral to *Todea*, *Leptopteris* and *Osmundopsis*’. Our clock-based analyses do not support the close relationship between *Osmundopsis* and *Todites*, instead placing *Osmundopsis* within an osmundoid clade (Fig. 1; Supplementary Data Fig. S2). However, this result is sensitive to methodological choices. Under implied weighting parsimony (k10–k15), *Osmundopsis* is rendered as sister to *Cacumen* and the leptopteroids (Supplementary Data Fig. S1). In non-clock BI, *Osmundopsis*, *Cacumen* and *Todites* form a monophyletic group nested within leptopteroids (Supplementary Data Fig. S1). Conversely, a sister relationship between *Todites* and a clade comprising *Todea*, *Leptopteris* and *Osmundopsis* is not supported across our analyses (Supplementary Data Figs S1 and S2). Thus, although the assumption of a clock rejects the hypotheses of Phipps *et al.* (1998), our non-clock inferences are consistent with Phipps *et al.* (1998) by supporting a close relationship among *Osmundopsis*, *Todites* and the leptopteroids.

Our age estimates for crown osmundoids and crown leptopteroids differ by ~10 and ~20 Mya, respectively, from the median ages previously inferred in cFBD and TED analyses (Grimm *et al.*, 2015). Differences in node age estimates and phylogenetic affinities relative to previous studies can be attributed to three, non-mutually exclusive, methodological aspects: fossil sampling; lack of constrained node ages; and tree models. First, previous cFBD and TED analyses were based on datasets that scored permineralized stems (Grimm *et al.*, 2015). In contrast, our morphological dataset sought to enhance the sampling of leptopteroid fertile fronds. These discrepancies in fossil sampling might partly account for differences in the inferred topologies (e.g. Saladin *et al.*, 2017; Mongiardino Koch *et al.*, 2021). Second, previous cFBD analyses constrained the split between *Osmundastrum* and *Osmunda* based on *Osmunda claytoniites* Phipps *et al.* (Grimm *et al.*, 2015). Such a constraint is conservative because it does not assume an *O. claytoniites*–extant *Osmunda* relationship. However, it assumes that leptopteroids are sister to *Osmundastrum* + crown osmundoids, which conflicts with molecular and fossil evidence (Metzgar *et al.*, 2008; Carvalho *et al.*, 2013). Instead of constraining topologies to infer node ages, our analyses relied on the combined data and the model to infer divergence times.

Finally, for Osmundaceae, uniform tree priors have been used in TED analyses, whereas cFBD processes have been used for node-constrained molecular dating (Grimm *et al.*, 2015). Although fossil sampling may influence clock analyses, major differences in node ages commonly originate from alternative tree models (e.g. May *et al.*, 2021). Relative to sFBD models that allow shifts in diversification rates through time, uniform and cFBD tree priors are known to yield (often unrealistic) old ages (May *et al.*, 2021; Zhang *et al.*, 2023). Furthermore, as opposed to the

cFBD model, sFBD has proved robust to violations of sampling assumptions in TED analyses (Zhang *et al.*, 2023). As in the analyses by May *et al.* (2021), our inferences implemented tree models where diversification (and fossilization) rates could vary through time. Although extensively evaluating tree models is beyond the scope of our study, the difference observed between our sFBD analyses and previous inferences can reasonably be attributed to the impact of alternative models.

High extinction rates and morphospace contraction amid global climatic events

Our best-fitting model indicates that net diversification rates were low between the TOA and the initial opening of the Drake Passage (193–49 Mya; TBii; Fig. 1B). Conversely, net diversification rates were higher early in the history of Osmundaceae (Tbi) and again by the onset of Drake Passage and the Middle Miocene Climate Transition (TBiv; Fig. 1B). In contrast, relative extinction (turnover) rose sharply in the interval 193–49 Mya (TBii; Fig. 1B). That is, the interval 193–49 Mya (TBii) was characterized by high extinction (low net diversification) and high species replacement (turnover), where lineages persisted for short periods. In fact, by the end of the TOA event, over half of the lineages went extinct (Fig. 1C). This suggests a mechanism in which lineages accumulated during climatically mild intervals and declined during rapid warming (Fig. 1C). During the TOA event, fossil floras are known to have transitioned from mesophytic to xerophytic assemblages (Slater *et al.*, 2019). Although some post-TOA floras recovered rapidly afterwards (García Joral *et al.*, 2011), our results indicate that high extinction rates in Osmundaceae persisted throughout the Mesozoic, delaying recovery until the Miocene (Fig. 1B, C).

Consistent with the loss of lineages, our DTT analyses inferred a reduction in morphospace size after the TOA event (Fig. 2B). The ‘general DTT trend’, wherein clades reach their maximum disparity early in their history (Hughes *et al.*, 2013), is supported by our results (Fig. 2B). Morphospace ‘density’ and ‘displacement’ remained stable in our analyses (Fig. 2B), which agrees with post-TOA morphotypes persisting in their morphospace configuration (Matamoro-Vidal *et al.*, 2016). Although the link between disparity and extinction remains debated (Polly, 2023), simulations indicate that a simultaneous decline in both LTT and DTT metrics signals a selective extinction event (Puttick *et al.*, 2020). Therefore, the concomitant low diversification rates, lineage loss and morphospace contraction observed here suggest selective extinction triggered by the TOA event (Figs 1 and 2; Puttick *et al.*, 2020).

Our best-fitting model also recovers character change rates exceeding the upper 95 % CI until the end of the Early Jurassic (~175 Mya; Fig. 2A). Except for binary fertile characters in crown osmundoids, character change rates after the Early Jurassic have either been slower than the lower 95 % CI or fallen within the lower-upper 95 % CI range (Fig. 2A; Supplementary Data Fig. S6). This pattern of post-Early Jurassic decelerated rates is coeval with the long-term extinction interval after the TOA event and is consistent with Mesozoic floristic transitions from mesophytic to

xerophytic floras (Fig. 2B; Slater *et al.*, 2019). Despite being a high-extinction interval, our clock phylogeny infers the radiation of crown osmundeoids and crown leptopteroids in TBii (Fig. 1A), indicating that high extinction did not completely preclude diversification in Osmundaceae. Consistent with a mechanism in which speciation is promoted under mild temperatures (Fig. 1C), crown osmundeoid and crown leptopteroid morphotypes might have been favoured when temperatures dropped ~ 49 Mya, resulting in lineages that diverged morphologically (Fig. 2C).

An Antarctic cradle for crown leptopteroids

Our biogeographical inferences imply that crown leptopteroids did not radiate via dispersal through Antarctica but instead radiated from Antarctic populations (Fig. 3). The broad distribution observed in some extant Osmundaceae is contingent on a once widespread Pangean population, such that the extant taxa could be seen as its remnants (Fig. 3A–C; Supplementary Data Fig. S6). The ancestor of crown leptopteroids migrated from northern latitudes to an ice-free Antarctica

(Fig. 3D–F). Although this migration might seem consistent with the classic view of Antarctica as a corridor (e.g. Winkworth *et al.*, 2015), the most recent common ancestor of crown leptopteroids was widespread across Antarctica at the time *Todea* and *Leptopteris* radiated (Fig. 3F). Thus, our results place crown leptopteroids as relicts of a once widespread Antarctic population.

Our analyses expand on the existing evidence that positions Antarctica as a main centre of distribution for past plant communities (Escapa *et al.*, 2011; Cantrill and Poole, 2012). Based on the frond record from Antarctic deposits, ferns are recognized as a dominant component of past Antarctic floras (Cantrill and Nagalingum, 2005; Iglesias, 2016; Iglesias *et al.*, 2025; Silva *et al.*, 2025). Yet, the exact biogeographical processes associated with Antarctic floras are unclear (Wilf *et al.*, 2013). In our analyses, low diversification rates and high relative extinction (turnover) rates coincided with a decline in lineage number and morphospace size contraction between 193 and 49 Mya (Figs 2 and 3). These inferences, therefore, suggest that crown leptopteroids speciated from a morphologically depauperate Antarctic population.

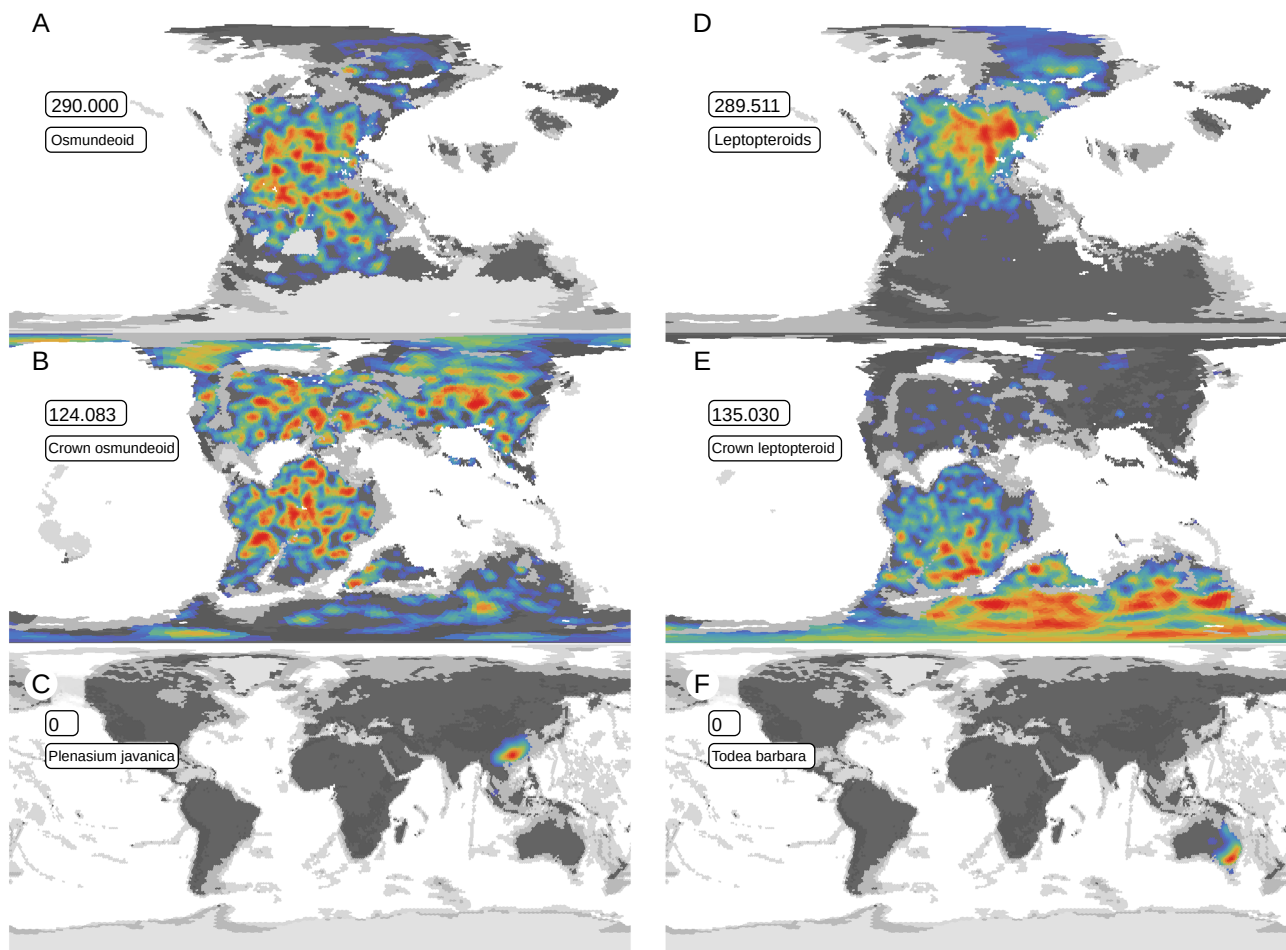


FIG. 3. Biogeographical history for selected clades of Osmundaceae: (A) osmundeoids (clade II; Fig. 1A); (B) crown osmundeoids (clade VII; Fig. 1A); (C) *Plenasium javanicum*; (D) leptopteroids (clade III; Fig. 1A); (E) crown leptopteroids (clade IV; Fig. 1A); and (F) *Todea barbara*. The maps show the palaeogeographical reconstruction at a given time (indicated in million years ago), and the shading reflects the ancestral pixel probability (highest probability in red, lowest in blue) as given by the 95 % cumulative kernel density function.

Our findings likewise suggest a novel mechanism to account for the radiation of extant Osmundaceae. The description of the Patagonian *T. amissa* led to the hypothesis that Asian tropical forests might have acted as refugia for the subsequent diversification of extant leptopteroids (Carvalho *et al.*, 2013). The diffusion models reconstructed herein suggest an alternative scenario wherein crown osmundeoids were widespread across Laurasia and crown leptopteroids across Antarctica before the radiation of Asian/Oceanic species (e.g. *Plenasium javanicum* and *Todea barbara*; Fig. 3; Supplementary Data Videos S2–S6). That is, extant leptopteroids diversified from an Antarctic source population following the drift of Antarctica and the onset of the Middle Miocene Climate Transition (Fig. 3D, F; Supplementary Data Videos S5 and S6). Hence, instead of radiating from Asian tropical refugia, our inferences imply that crown leptopteroids radiated from an Antarctic ancestral population.

Conclusion

We present the first dated phylogeny focused on fertile leptopteroids under a skyline FBD model that allows diversification and fossilization rates to shift across time bins. Consistent with the leptopteroid fossil record, our analyses indicate that leptopteroid diversity peaked before the Early Jurassic (Fig. 1A, B). This was followed by a prolonged interval of high extinction until the Miocene, resulting in a sharp decline in the number of lineages (Fig. 1C). Such long-term loss of lineages was accompanied by a deceleration in character change rates and morphospace contraction (Fig. 2). Deceleration in character change rates and morphospace contraction are coeval with major Mesozoic vegetation turnovers, in which mesophytic floras shifted to xerophytic ecosystems (Krassilov, 1981; Slater *et al.*, 2019).

Although this advances our understanding of Osmundaceae evolution, an important caveat remains when interpreting Osmundaceae phylogeny. Osmundaceae phylogenetics has relied heavily on permineralized fossils (Miller, 1971; Jud *et al.*, 2008; Tian *et al.*, 2008, 2014; Bomfleur *et al.*, 2017). However, these phylogenies have often differed from those inferred from fertile fronds (Carvalho *et al.*, 2013; Grimm *et al.*, 2015; Bomfleur *et al.*, 2017; Urrea *et al.*, 2025). Although relationships among extant taxa have remained consistent across studies, divergence times inferred here differ from those obtained in TED analyses that included permineralized axes or cFBD models (Grimm *et al.*, 2015; Table 1), highlighting the impact of the underlying model.

It has been argued that misplaced fossils do not lead to substantial errors in divergence time estimates under cFBD dating (Barido-Sottani *et al.*, 2023; Mongiardino Koch *et al.*, 2023). One might expect to infer divergence times in fertile Osmundaceae by constraining nodes with permineralized fossils while accommodating uncertainty through broad priors (Barido-Sottani *et al.*, 2023). In fact, such an approach has been adopted in previous analyses, where cFBD-based phylogenies of extant taxa were constrained with both permineralized axes and fronds (Grimm *et al.*, 2015). However, besides uncertainty, Osmundaceae fossils pose another challenge: permineralized species of different ages may correspond to the same

frond species and *vice versa*, which can potentially extend the age of the putative fossil organism (Bomfleur *et al.*, 2017). Originally framed as an empirical demonstration of character dependence, the study by Urrea *et al.* (2025) further compounded this issue by rendering some permineralized fossil species as non-Osmundaceae. Further progress should help to clarify the correspondence between permineralized and frond species and extensively assess the impact of model choice in total-evidence analyses of Osmundaceae.

SUPPLEMENTARY DATA

Supplementary data are available at *Annals of Botany* online and consist of the following. Table S1: extant taxon and fossils included in the analyses, their stratigraphic range, specimen number, repository, literature of reference, and accession numbers for the molecular data (*atpA*, *rbcl*, *rbcl-accD*, *rbcl-atpB*, *rps4-trnS*, *trnG-trnR* and *trnL-trnF*). Dashes (‘–’) indicate non-applicable information or no available sequence. Note that, for some fossils, the type specimen is often the illustration provided in the original description or amendment. Likewise, the final repository is unknown for many of the older fossils (indicated with a question mark). In these cases, a reference specimen other than the holotype is given. Table S2: fit comparison between alternative models. Average marginal likelihoods and 2ln Bayes factors calculated from stepping-stone analyses. ‘Source partitioning’ and ‘state partitioning’ refer to the alternative partitioning schemes for the morphological dataset. ‘Random’ and ‘diversity’ refer to strategies for sampling extant taxa in the sFBD process. Alternative clock models are indicated with ‘IGR’ and ‘TK02’, respectively. The best-fitting model (M2: ‘source partitioning’ + IGR + ‘random’) is highlighted in bold. Table S3: distance and speed metrics as inferred in PhyGeo, using the dated phylogeny under the best-fitting model and georeferenced points for the included fossil and extant taxa. ‘Slower’ and ‘faster’ refer to the proportion of particles that were faster and slower than 95 and 5 %, respectively, of simulated particles (for details, see Arias, 2024). Node numbers refer to those of Supplementary Data Fig. S4. Figure S1: topologies inferred under implied-weighting parsimony and non-clock Bayesian inference. Figure S2: topologies inferred under clock Bayesian inference across implemented models. Figure S3: disparity through time for different partitioning schemes of the morphological dataset: vegetative and fertile, nominal and ordinal characters. Figure S4: dated phylogeny inferred under the best-fitting model. Nodes discussed in the main text and Supplementary Data Table S3 are numbered for reference. Figure S5: georeferenced points for extant and fossil Osmundaceae. Figure S6: dated phylogeny, with branches coloured according to change rates for the ‘vegetative partition with four character states’. Branches are coloured as in Fig. 2: ‘slower than the lower 95 % CI’ (light blue); ‘within lower and upper 95 % CI’ (grey); and ‘faster than the upper 95 % CI’ (red). Main groups of interest are indicated as: ‘SO’ (stem osmundeoids); ‘CO’ (crown osmundeoids); ‘SL’ (stem leptopteroids); and ‘CL’ (crown leptopteroids). Appendix S1: list of morphological characters. Note that both the morphological and the combined datasets

are readily available in computer-readable format from Morphobank (project 6170). **Video S1:** video showing the trajectory of the most recent common ancestor of the osmundoid clade (clade 3 from [Supplementary Data Fig. S4](#)). **Video S2:** video showing the trajectory of the most recent common ancestor of *Osmunda* (clade 26 from [Supplementary Data Fig. S4](#)). **Video S3:** video showing the trajectory of the most recent common ancestor of *Plenasium* (clade 31 from [Supplementary Data Fig. S4](#)). **Video S4:** video showing the trajectory of the most recent common ancestor of crown leptopteroids (clade 68 from [Supplementary Data Fig. S4](#)). **Video S5:** video showing the trajectory of the most recent common ancestor of extant *Leptopteris* (clade 74 from [Supplementary Data Fig. S4](#)). **Video S6:** video showing the trajectory of the most recent common ancestor of extant *Todea* (clade 71 from [Supplementary Data Fig. S4](#)).

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CONFLICT OF INTEREST

None declared.

AI ASSISTANCE

AI was not used to conduct this study or write the manuscript.

AUTHOR CONTRIBUTIONS

M.J.U. compiled the morphological dataset with the assistance of A.Y. and ran the parsimony analyses. J.R.F. conceptualized the study, ran the Bayesian and biogeographical analyses and wrote the initial manuscript with input from the other authors.

DATA AVAILABILITY

The morphological dataset is freely available from MorphoBank, project 6170. Scripts to perform phylogenetic and downstream analyses are available from GitHub: <https://github.com/jorgerfl/osmunda26>. The combined and morphological dataset, including character definitions, is also available from MorphoBank (project 6170).

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